# CITS4404 Project — Building AI Trading Bots: Nature-Inspired Optimisation Algorithms Implementation and Experimentation, and Evaluation

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2755 words

## I. ALGORITHMS INVESTIGATED

Six optimisation algorithms were evaluated across all experimental configurations, spanning four structural categories: swarm-based, physics-inspired, ecosystem-inspired, and classical local search.

Particle Swarm Optimization, originally proposed by Kennedy and Eberhart [1] and extended with an inertia weight by Shi and Eberhart [2], is a swarm method in which each particle maintains both a personal best position and a shared global best. The inertia weight $w$ governs the balance between exploration and exploitation through the velocity update rule. Atomic Orbital Search (AOS) [3] is a physics-inspired method detailed in Section III; it employs shell-based clustering around a nucleus (global best) as its primary exploration mechanism, without per-agent personal memory. African Buffalo Optimization (ABO) [4] encodes two herd vocalisations — an exploratory foraging call and an exploitative regrouping call — and like PSO retains both personal and global best positions. Manta Ray Foraging Optimization (MRFO) [5] models three foraging behaviours (chain, cyclone, and somersault foraging) but maintains only a global best, with no per-agent memory. Symbiotic Organisms Search (SOS) [6] simulates mutualistic, commensal, and parasitic ecological interactions, applying all three phases to every organism per iteration; this yields three fitness evaluations per organism per iteration and requires no algorithm-specific hyperparameters. Generalized Pattern Search (GPS) [7] is a classical single-state local optimiser that evaluates a fixed pattern of directions from the current best solution, with no population and no historical memory beyond the current position; it serves as a deliberate baseline to quantify the value of population-based diversity.

Table I summarises the distinguishing characteristics of each algorithm.

TABLE I. Summary of the six optimisation algorithms evaluated. Memory column describes which historical positions are retained per agent. Eval/iter denotes fitness evaluations per agent per iteration.

| Algorithm | Year | Category | Population-based | Memory | Eval/iter |
|---|---|---|---|---|---|
| AOS | 2021 | Physics | Yes | Global (nucleus) | 1 |
| PSO | 1998 | Swarm | Yes | Personal + Global | 1 |
| MRFO | 2020 | Swarm | Yes | Global only | 1 |
| SOS | 2014 | Ecosystem | Yes | Global only | 3 |
| ABO | 2015 | Swarm | Yes | Personal + Global | 1 |
| GPS | — | Local search | No | Current best only | 1 |

## II. BOT DESIGN AND PARAMETRISATION

### A. Trading Signal and Fitness

Each strategy produces a short-window (low-period) and a long-window (high-period) smoothed price series. The trading signal detects crossover events in their difference using a two-point weighted kernel:

$$d_t = \text{Series}_{\text{short},t} - \text{Series}_{\text{long},t} \quad , \quad C_t = \tfrac{1}{2}(\text{sign}(d_t) - \text{sign}(d_{t-1})) \tag{1}$$

$$S = \begin{cases} \text{buy} & \text{if } C_t = -1 \text{ (short crosses below long)} \\ \text{sell} & \text{if } C_t = +1 \text{ (short crosses above long)} \end{cases} \tag{2}$$

where $d_t$ is the difference of the two smoothed series at time $t$, $C_t \in \{-1, 0, +1\}$ is the crossover indicator, and $S$ is the resulting trade action. The signal is non-zero only at the moment of a crossover; it is zero on all other timesteps.

The fitness function measures final USD holdings after simulating trades over the full training price series:

$$\text{Fitness}(X) = \text{USD}_{\text{final}} \tag{3}$$

where $X$ is the candidate parameter vector, $\text{USD}_{\text{final}}$ is the cash held at the end of the simulation, and the starting balance is $\text{USD}_{\text{initial}} = \$1000$. A transaction fee of 3% is applied to each trade. The simulation is a single continuous pass over the training data with no mid-run resets.

One acknowledged limitation is that the fitness function imposes no penalty for non-trading: if no crossover ever occurs, $C_t = 0$ for all $t$ and the bot holds cash throughout, returning $\text{Fitness}(X) = \text{USD}_{\text{initial}}$. This is a valid but uninformative fixed point. The degenerate optimum was observed empirically — MRFO converged to parameter regions with no crossover signal under certain configurations. A risk-adjusted metric such as the Sharpe ratio would penalise this behaviour but was not adopted here; the limitation is documented rather than silently corrected.

### B. Strategy Configurations

Five strategy configurations define the hypothesis space presented to each optimiser.

*Strategy 1 — Double SMA crossover ($d = 2$).* Both the short window $x_0 \in [5, 50]$ and the long window $x_1 \in [51, 200]$ are optimised. The non-overlapping constraint (short < long) is guaranteed by construction through the parameter bounds, with no runtime enforcement required. The SMA assigns uniform weight to all prices within the window, making it insensitive to the ordering of recent versus older prices.

*Strategy 2 — Double LMA crossover ($d = 2$).* The parameter structure is identical to Strategy 1 ($x_0 \in [5, 50]$, $x_1 \in [51, 200]$). The LMA assigns linearly decaying weights so that more recent prices contribute more than older ones, providing momentum sensitivity that the uniform SMA cannot capture. The optimiser thus faces the same two-dimensional search space as Strategy 1 but with a smoother series that responds differently to regime changes.

*Strategy 3 — EMA with shared $\alpha$ ($d = 3$).* A short span, a long span, and a single shared smoothing parameter $\alpha \in (0, 1)$ are optimised jointly. Sharing one $\alpha$ value across both EMAs constrains both series to decay at identical rates; the frequency contrast that drives the crossover signal therefore depends entirely on the span difference rather than on independent decay dynamics. This effectively collapses the degree of freedom to approximately 2.5 effective dimensions in practice.

*Strategy 4 — EMA with independent $\alpha$ ($d = 7$).* Short span, long span, $\alpha_{\text{short}}$, and $\alpha_{\text{long}}$ are each optimised independently, along with additional EMA window parameters. Decoupling the smoothing parameters gives the short-term EMA its own reactivity and the long-term EMA its own stability, restoring the frequency contrast that shared $\alpha$ eliminates. The expanded search space ($d = 7$ versus $d = 3$) carries a higher exploration cost but provides genuinely greater expressiveness: the two EMAs can now operate at qualitatively different timescales simultaneously. Empirically, this configuration produced the highest Kruskal-Wallis $H$-statistics across all strategy conditions, suggesting that the larger search space amplifies inter-algorithm performance differences rather than masking them.

*Strategy 5 — Weighted MA combination ($d = 14$).* SMA, LMA, and EMA signals are combined as a weighted sum with learned weights $w_1$, $w_2$, $w_3$. All three weights are treated as free optimisation dimensions and normalised at evaluation time (each $w_i$ divided by $w_1 + w_2 + w_3$), yielding a 14-dimensional parameter vector that includes window parameters for each MA type plus EMA spans and smoothing factors. Deriving $w_3 = 1 - w_1 - w_2$, as the project brief implies, would create asymmetric optimisation pressure because $w_3$ is never directly searched; retaining all three as free variables ensures each weight receives equal search effort. The 14-dimensional space is the largest tested, and the curse of dimensionality is a genuine concern when each optimiser is limited to 30 runs.

### C. Justified Deviations from Project Brief

TABLE II. Summary of implementation decisions that deviate from the project brief, with justification for each.

| Deviation | Brief approach | Our implementation | Justification |
|---|---|---|---|
| EMA normalisation | Optional | Enabled by default | Unnormalised kernels produce span-dependent amplitude distortion, making fitness values non-comparable across parameter configurations |
| $d = 14$ weight derivation | $w_3$ derived as $1 - w_1 - w_2$ | All three weights free; normalised at eval time | Derived $w_3$ receives no direct search pressure, creating asymmetric optimisation bias |
| $n_{\text{shells}}$ assignment | Fixed shell count | $\min(n_{\text{shells}}, \lvert\text{remaining}\rvert)$ | Prevents degenerate empty shell states in high-dimensional runs where population may be smaller than the configured shell count |
| EMA $\alpha$ across spans | Shared $\alpha$ (Strategy 3) | Independent $\alpha_{\text{short}}$, $\alpha_{\text{long}}$ (Strategy 4) | Shared $\alpha$ constrains both series to identical decay rates, eliminating the frequency contrast that makes the crossover signal informative |

## III. ALGORITHM SELECTION

All six algorithms described in Section I were evaluated. This section explains the rationale behind each inclusion and details the primary algorithm, AOS.

*GPS as a deliberate baseline.* The central empirical question is whether population-based global search is necessary on a moving-average (MA) parameter fitness landscape, or whether a single-state local optimiser is sufficient. GPS [7] has no population, no escape mechanism from local optima, and no memory of previously evaluated positions beyond the current best. Including it provides a direct lower bound: if GPS performs competitively, population diversity contributes little; if GPS underperforms consistently, the landscape demands it.

*PSO as a population-based benchmark.* PSO [2] is one of the most extensively studied metaheuristics and provides a reliable population-based performance reference. It is also structurally related to AOS — both use a global nucleus/best to guide the population — making a paired comparison directly informative about the contribution of personal best memory. If AOS and PSO are statistically indistinguishable, per-particle historical memory may or may not add marginal benefit in this problem domain.

*AOS as the physics-based algorithm compared to the others being biology-based.* The MA parameter fitness landscape is expected to be multimodal: distinct parameter combinations can produce similar short-term profits through qualitatively different mechanisms, creating multiple local peaks of comparable height. AOS [3] was selected because its shell-based population structure explicitly supports simultaneous exploitation of multiple landscape regions. Candidate solutions (electrons) are assigned to concentric shells around the nucleus (current global best) according to their fitness rank; electrons in inner shells exploit promising regions while outer-shell electrons conduct broader exploration. Transitions between shells are governed by a quantum-mechanical photon absorption and emission analogy: an electron absorbs energy (moves to an outer shell, exploration) when its transition energy falls below the shell's binding energy, and emits energy (moves to an inner shell, exploitation) otherwise. This structured push-pull between shells provides more persistent population spread than PSO's convergent swarm behaviour, which collapses toward the global best over time. AOS was published in 2021 and has not, to our knowledge, been applied to financial parameter optimisation in the literature.

The pseudocode below summarises the core AOS iteration:

```
Initialise N electrons randomly in the search space
Set nucleus := position of the best electron (highest fitness)
Assign electrons to K shells by fitness rank (binding energy)

FOR each iteration t = 1, 2, ..., T:
    FOR each shell k = 1, ..., K:
        Compute shell binding energy E_k
        FOR each electron i in shell k:
            Sample transition energy E
            IF E < E_k  (electron absorbs energy):
                Move to outer shell; apply exploration step
            ELSE  (electron emits energy):
                Move to inner shell; apply exploitation step
```

```
            Update position via quantum transition equation
            Evaluate fitness of new position
    Apply LE perturbation: randomly displace the
        lowest-energy electron to escape local optima
    Update nucleus if any electron improves best fitness

RETURN nucleus position as best solution found
```

*LE degradation — an observed weakness.* The Lowest Energy (LE) perturbation step is designed to prevent stagnation by randomly displacing the electron with the worst fitness. In practice, the LE designation is not stable: as other electrons improve their fitness across iterations, the electron currently labelled LE may no longer occupy the position most in need of escape relative to the updated nucleus. The original paper [3] does not address this drift. This was observed empirically across runs and is logged per iteration; it is documented here rather than patched, as any correction would constitute a modification to the published algorithm.

*Metaparameter choices.* Population size was scaled with dimensionality — larger populations were used for the $d = 14$ weighted strategy to partially offset the curse of dimensionality. The shell count $n_{\text{shells}}$ is implemented as an upper cap rather than a fixed assignment: each iteration uses $\min(n_{\text{shells}}, |\text{remaining population}|)$ shells to prevent degenerate empty-shell states that can arise in high-dimensional runs where effective population diversity is reduced. The iteration budget was fixed uniformly across all six algorithms to ensure a controlled comparison of per-iteration effectiveness.

## IV. EXPERIMENTS AND EVALUATION

### A. Dataset

Daily BTC/USD OHLCV data sourced from Kaggle spanning 2014 to 2022 was used. The training set covers the pre-2020 period (approximately 2014–2019) and the test set covers 2020–2022. The split was not chosen arbitrarily: post-2020 data encompasses the COVID-19 market crash of March 2020, the 2021 bull run, and the 2022 bear market, constituting three qualitatively distinct market regimes. This is an intentional out-of-distribution stress test rather than a random holdout. Its purpose is to measure whether parameter configurations optimised on one regime generalise when the underlying market dynamics shift substantially — a more meaningful evaluation of strategy robustness than a matched-regime random split would provide.

### B. Experimental Design

The full factorial design comprised 6 algorithms $\times$ 5 strategies $\times$ 2 initialisation modes $\times$ 30 runs = 1800 total trials. Two initialisation modes were tested: fixed initialisation, in which all runs begin from a deterministic seed position, and random initialisation, in which each run begins from a position drawn uniformly at random from the search space. The paired comparison of fixed versus random initialisation tests sensitivity to starting conditions. Strong dependence on initialisation would indicate algorithmic fragility, with convergence driven by the seed rather than by the search mechanism. A run count of 30 was chosen as a practical compromise between statistical power and the total computational cost of 1800 trials; it is sufficient to apply non-parametric distributional tests but modest relative to what would be preferred for high-variance optimisers.

### C. Statistical Analysis

The Kruskal-Wallis $H$-test was used in preference to one-way ANOVA because the structural properties of optimiser output systematically violate the normality assumption required by ANOVA. Optimisers can produce runs that converge to the degenerate no-trade optimum (fitness near $1000, the starting balance), occasional jackpot runs that far exceed the typical distribution, and heavy-tailed spread in between. These outliers are not noise — they reflect real behavioural modes of the algorithms — and would distort ANOVA $F$-statistics. Kruskal-Wallis operates on ranks and is robust to such distributional shapes.

For each of the ten strategy $\times$ initialisation mode combinations, the Kruskal-Wallis test was applied with algorithm as the factor variable, holding strategy and initialisation constant. This isolates the algorithm effect from strategy-level confounds: testing across all strategies simultaneously would conflate the variance introduced by different fitness landscape shapes with the variance attributable to algorithmic differences. Pairwise post-hoc comparisons used Dunn's test with Bonferroni correction. A significance threshold of $\alpha = 0.05$ was applied throughout.

## V. RESULTS AND ANALYSIS

### A. Training Results


FIG. 1. Dunn's post-hoc pairwise $p$-values (Bonferroni-corrected) for all strategy × initialisation combinations — training fitness. Red indicates significant difference ($p < 0.05$); green indicates no significant difference.

The Kruskal-Wallis test was significant across all ten strategy × initialisation combinations ($p < 10^{-20}$ in every case), with $H$-statistics ranging from 106.2 (LMA, random initialisation) to 151.7 (EMA-independent, fixed initialisation). Algorithm choice has a universally significant effect on training fitness. The `ema_independent` strategy consistently yielded the highest $H$-statistics, suggesting that the expanded seven-dimensional search space amplifies inter-algorithm performance differences more than the two-dimensional strategies do — algorithms with stronger exploration mechanisms extract more from the additional degrees of freedom.

Four patterns emerge from Dunn's pairwise post-hoc comparisons, illustrated in Figure 1. First, GPS is consistently and significantly worse than every other algorithm across all conditions. This confirms that single-state local search cannot reliably navigate the multimodal MA fitness landscape: without population diversity, GPS converges to the nearest local optimum from its starting position and cannot escape. Second, PSO and AOS are not significantly different in the majority of strategy $\times$ initialisation pairs. Despite AOS lacking per-particle personal memory, its shell-based exploration achieves comparable training fitness to PSO, suggesting that personal best tracking contributes little marginal benefit when global best guidance is combined with structured shell diversity. Third, PSO and SOS are also statistically equivalent in most training conditions. SOS requires three fitness evaluations per organism per iteration — a direct consequence of the mutualism, commensalism, and parasitism phases — while PSO requires one. Statistical equivalence under matched-iteration budgets implies that a compute-normalised comparison (equalising the total number of fitness evaluations rather than the number of iterations) would likely favour PSO. Fourth, MRFO and ABO occupy intermediate positions that vary by strategy, with no consistent dominance pattern relative to the PSO-AOS-SOS cluster.

## B. Test Results

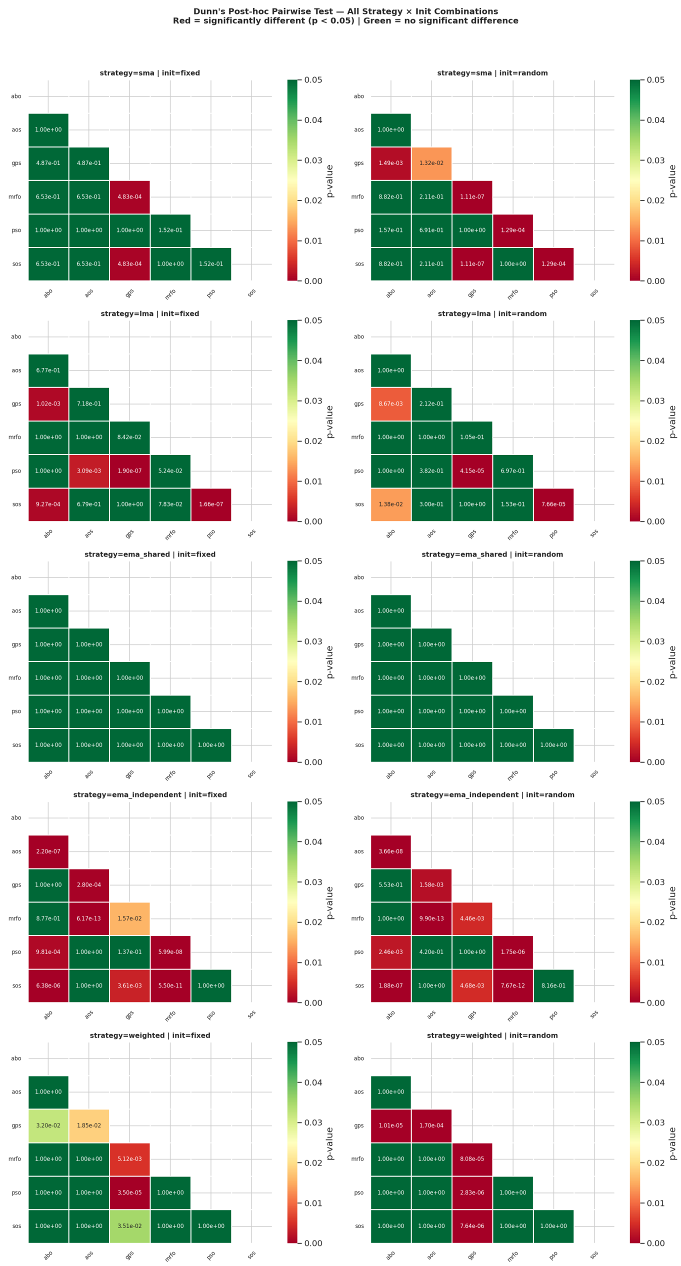


FIG. 2. Dunn's post-hoc pairwise $p$-values (Bonferroni-corrected) for all strategy $\times$ initialisation combinations — test fitness. Red indicates significant difference ($p < 0.05$); green indicates no significant difference.

On test data, the significant pairwise differences observed during training largely disappear (Figure 2). The most extreme case is `ema_shared`: under both fixed and random initialisation, every pairwise $p$-value equals 1.000, meaning the six algorithms produce statistically indistinguishable test fitness distributions under that strategy. The algorithms that were most differentiated during training — particularly the PSO-SOS-AOS cluster versus GPS — converge to similar test performance across strategies.

### C. Interpretation

The central finding is that training fitness differences did not generalise to test performance. Two non-exclusive explanations account for this. The first is overfitting to the training market regime: parameters optimised on the pre-2020 BTC market exploit structural patterns — momentum cycles, volatility regimes — specific to that period. The COVID-19 crash, the 2021 bull run, and the 2022 bear market introduce qualitatively different dynamics that these parameters are not positioned to exploit. The second explanation is that MA crossover signals are fundamentally weak predictors of BTC price movement across diverse regimes regardless of how precisely the parameters are tuned; if the strategy expressiveness is the binding constraint, no optimiser can recover transferable advantage from the parameter space. The fact that even the best-training algorithms fail to outperform on test data is consistent with the efficient market hypothesis for short-horizon technical signals on highly liquid assets. This is an empirical finding about the limits of MA-based strategy optimisation on non-stationary assets, not a failure of experimental design.

## VI. CONCLUSION

Algorithm choice has a universally significant effect on training fitness across all strategy and initialisation conditions, yet this advantage does not transfer to test performance. The binding constraint appears to be strategy expressiveness in the face of market non-stationarity rather than optimiser quality: once the target regime shifts, the parameters optimised on training data no longer generalise regardless of how effectively they were found.

AOS performs comparably to PSO despite lacking personal best memory. On this multimodal fitness landscape, global best guidance via nucleus tracking combined with shell-based population distribution appears sufficient; the additional per-particle history that PSO maintains provides no statistically significant advantage. This is a finding specific to the problem structure — the MA crossover landscape may not present the kind of elongated, directional ridges where personal memory typically helps.

GPS confirms that population diversity is necessary. Single-state local search is consistently and significantly worse than every population-based method across all conditions. The MA parameter landscape contains enough local optima that a deterministic pattern search from a fixed starting point cannot reliably reach high-fitness regions, establishing population-based global search as a structural requirement rather than an optional improvement.

SOS's three-evaluation-per-iteration cost is not justified by its performance differential over PSO. Because Dunn's test establishes distributional equivalence between PSO and SOS on training fitness under matched-iteration budgets, the threefold evaluation overhead buys no statistical advantage. A compute-normalised comparison equalising total fitness evaluations rather than iterations would be required to confirm this quantitatively, but the direction of the finding is clear.

Independent $\alpha$ in `ema_independent` provides genuinely greater expressiveness than the shared-$\alpha$ `ema_shared` configuration. The expanded search space produces more pronounced inter-algorithm differences (higher Kruskal-Wallis $H$-statistics), indicating that the additional degrees of freedom challenge the optimisers' exploration capabilities in a way that better reveals their structural differences.

The LE degradation in AOS is an empirically observed phenomenon. The lowest-energy electron's role as a local-optima escape mechanism degrades as the nucleus shifts across iterations and other electrons improve, because the LE designation tracks relative ranking within the current population rather than an absolute landscape property. This is a genuine weakness in the paper's formulation that warrants further investigation.

The no-trade degenerate optimum is an unpenalised artifact of using raw final cash as the fitness metric. A risk-adjusted metric such as the Sharpe ratio or return-per-trade would penalise conservative non-trading behaviour and remove this fixed point from the feasible optima.

*Outlook* Future work should adopt risk-adjusted fitness functions and regime-aware train/test splits that better reflect the non-stationarity of cryptocurrency markets. The train/test divergence observed here suggests the evaluation objective itself may require redesign: objectives that penalise in-sample overfitting — for example, through cross-validated segment fitness or walk-forward testing — would impose generalisation pressure that the current segment-average fitness does not provide.

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